

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
file_path = '/mnt/data/try.csv'

df = pd.read_csv('Family Income and Expenditure.csv')

X = df[['House Age']]
y = df['House Floor Area']

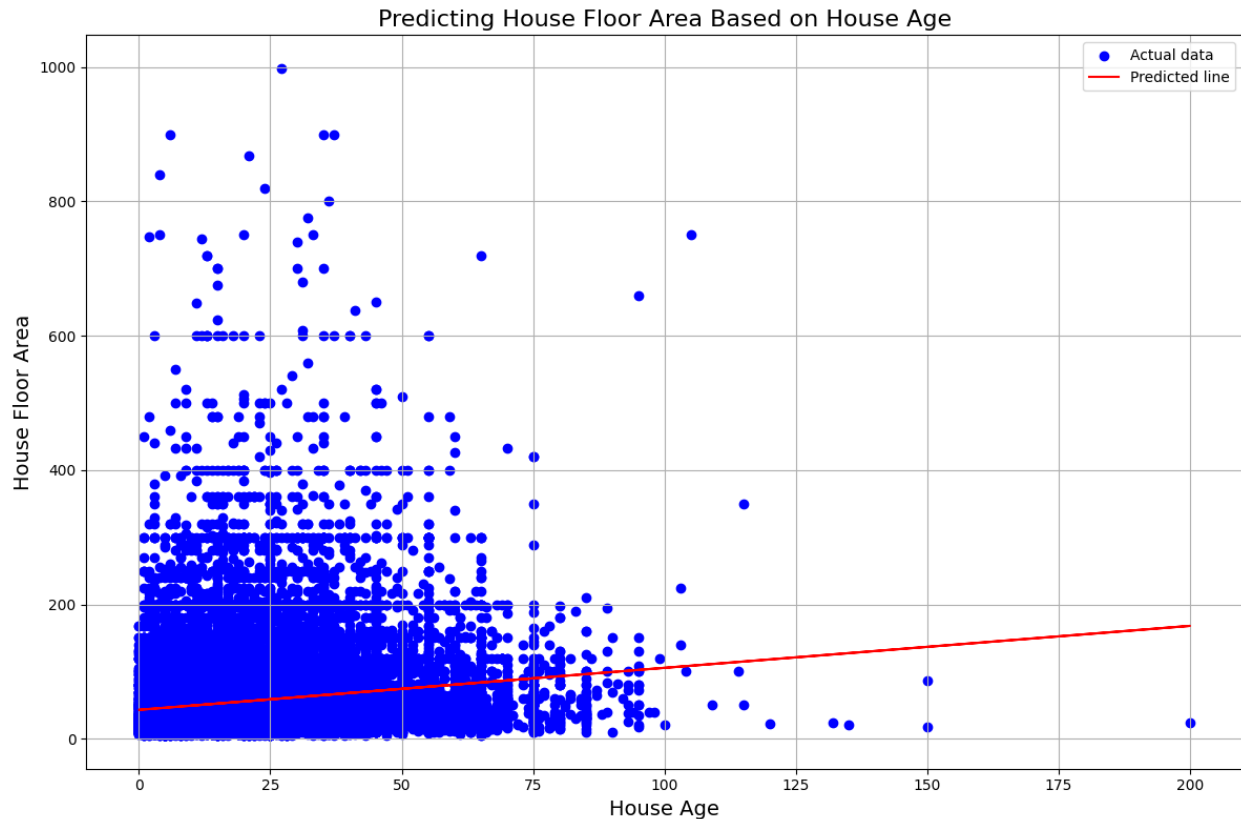
model = LinearRegression()

model.fit(X, y)

y_pred = model.predict(X)

plt.figure(figsize=(12, 8))
plt.scatter(df['House Age'], df['House Floor Area'], color='blue',
            label='Actual data')
plt.plot(df['House Age'], y_pred, color='red', label='Predicted line')

plt.title('Predicting House Floor Area Based on House Age',
          fontsize=16)
plt.xlabel('House Age', fontsize=14)
plt.ylabel('House Floor Area', fontsize=14)
plt.legend()
plt.grid(True)
plt.tight_layout()
plt.show()
```



```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
import matplotlib.pyplot as plt
import numpy as np

X = df[['Total Household Income', 'Education Expenditure']]
y = (df['Number of Personal Computer'] > 0).astype(int)

X_train, X_test, y_train, y_test = train_test_split(X, y,
test_size=0.3)

model = LogisticRegression(max_iter=300)
model.fit(X_train, y_train)

pred = model.predict(X_test)

unique, counts = np.unique(pred, return_counts=True)

plt.figure()
plt.pie(counts, labels=['No Computer', 'Has Computer'], autopct='%1.1f%%')
plt.title('Predicted Computer Ownership Distribution')
plt.show()
```

Predicted Computer Ownership Distribution

